

HCCL北极星平台 (HCCL-VM) 介绍

<https://gitee.com/cann>

CANN

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背景：大模型时代 HCCL 验证范式的全新演进

🚫 大模型时代物理验证的新命题

- **通信成瓶颈枢纽：**智算集群大模型（如 MoE、All-to-All）训练中，HCCL 的微小抖动在千卡规模会被指数级放大。
- **验证成本攀升：**物理上板验证依赖昂贵的集群资源，排队时间长，难以满足当前敏捷迭代的研发需求
- **全链路定界复杂：**集群中，软硬件耦合极深，快速区分是“算法逻辑 Bug”、“内存并发踩踏”还是“底层网络抖动”，极具挑战。

🚀 北极星验证新范式

秉持前置排雷理念，将必须上机才能发现的海量逻辑问题，拦截在纯软分析阶段，释放稀缺的真机算力潜能。

0 物理 NPU 占用

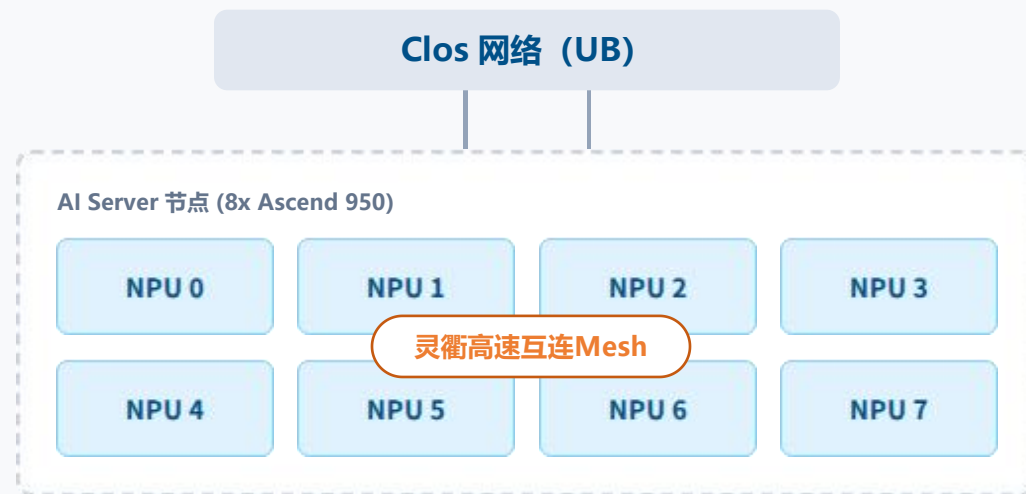
本地一键拉起

偶发转离线必现

<https://gitcode.com/cann>

代码仓：https://gitcode.com/cann/hcomm/tree/master/test/hccl_vm

典型 Ascend 950 多卡集群物理拓扑



⚠️ 硬件排障黑盒困境

多卡并发时，极小的算法逻辑越界极易被复杂的 **UB/ROCE** 等物理网卡与路由拓扑掩盖。传统模式需长周期 Dump 重放，定界效率极低。

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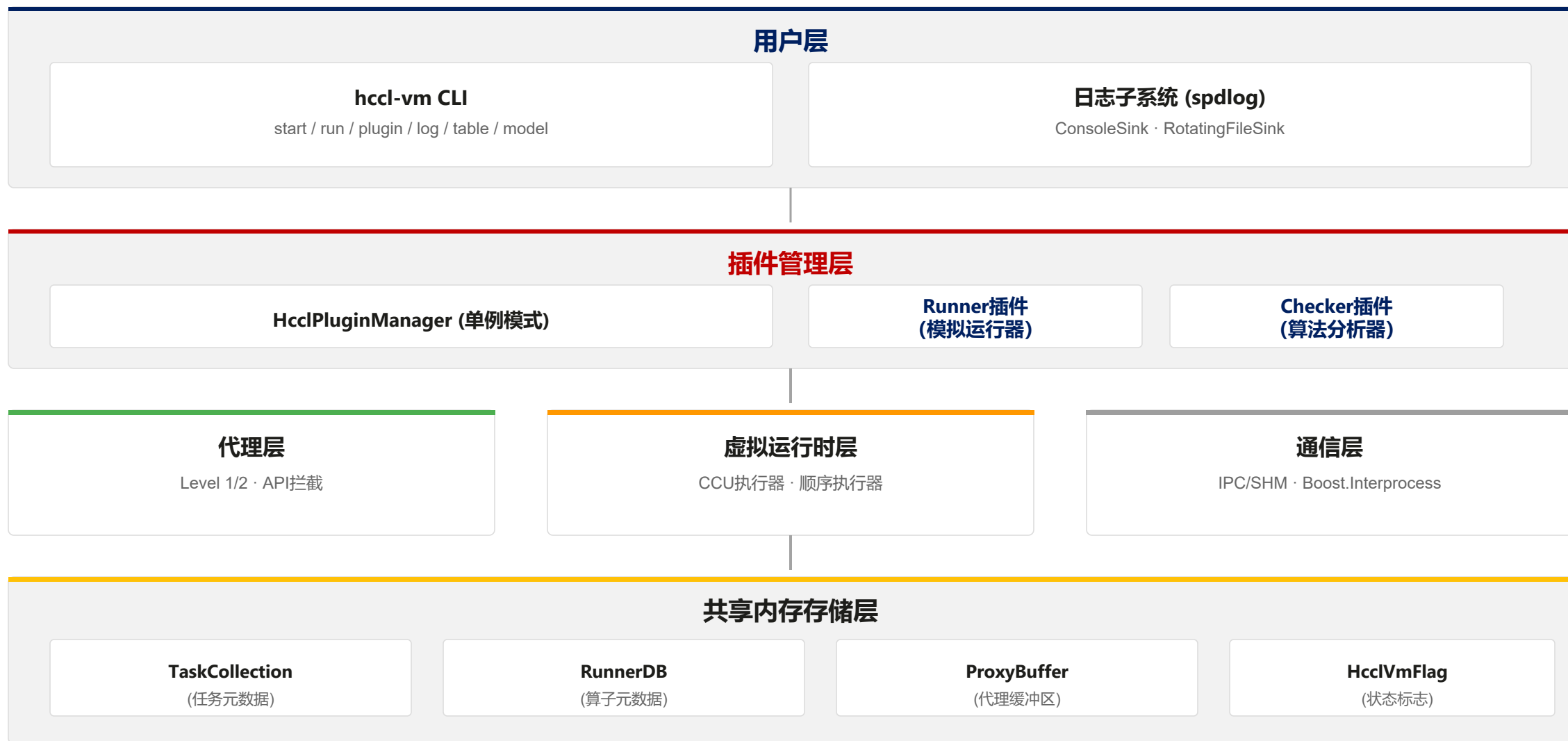
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北极星工具（HCCL-VM）：总体架构



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模拟运行器原理

核心机制

通过 LD_PRELOAD 动态库拦截技术，透明捕获 HCCL API 调用，提取任务元数据，在虚拟运行时中模拟执行并验证正确性。

执行流程



关键技术点

技术	实现方式	优势
LD_PRELOAD 动态库拦截	libhccl_proxy_level1/2.so	无侵入、透明拦截
共享内存	Boost.Interprocess	高效进程间通信
柔性数组	TaskCollection::tasks[]	可变数量任务存储
工厂模式	CcuExecutorFactory	易于扩展新指令类型

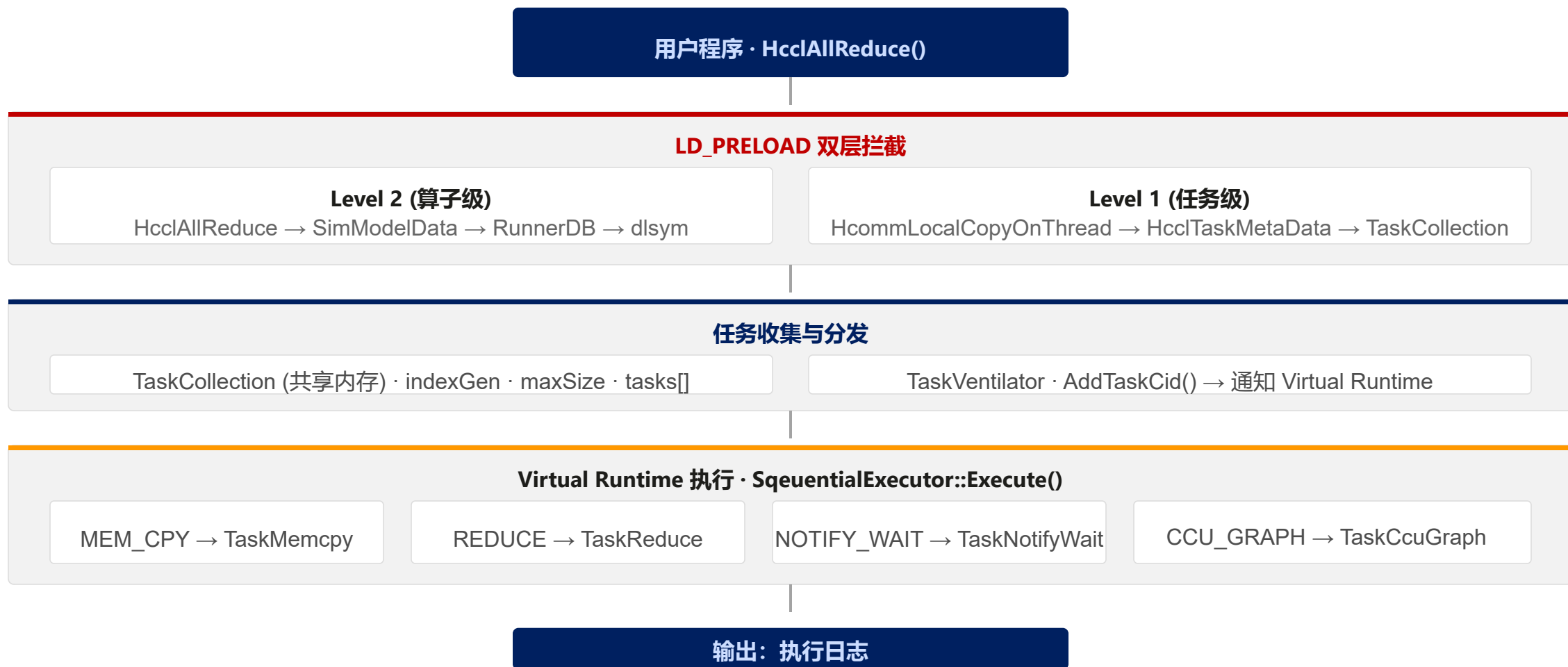
运行模式



<https://gitcode.com/cann>



模拟运行器：执行流程



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算法分析器原理

💡 核心洞察

- **动态时态捕获：**摒弃局限的静态源码推演，通过虚拟环境底层静默劫持，精准获取真实运行时的 API 调用序列。
- **自动化分析链路：**独创“输入捕获 → 拓扑成图 → 基础排雷 → 双路校验”的标准化、端到端验证流水线。

1

输入捕获

底层静默劫持真实 HCCL API，获取真实调用任务，重组为多组 Task Queues。

2

底座构建

提取通信与同步原语，建立 Post/Wait 配对与跨队列依赖，生成全局任务图。

3

前置排雷

执行基础结构校验：排查环形依赖、悬挂配对关系。回答“图能不能走通”。

4

双路高阶校验

支线 A (语义)：结果图案是否符合协议。
支线 B (内存)：执行中资源是否被踩坏。

🛡️ **设计优势：**系统关注指令因果关系和拓扑依赖，本质是对任务图做逻辑推演，而非真实申请内存去做物理执行。

算法分析器：语义校验引擎原理

💡 核心洞察

- **轻量化推演引擎：** 不需要进行真实浮点运算与物理周期仿真，通过数据结构定义不同任务，实现极低资源开销的逻辑推演。
- **血缘“染色”模型：** 跨越数值表象，通过追踪数据区间流向，高效且精确地验证最终输出图案与算子协议的一致性。

📦 动态传播规则

1. 覆盖式传播 (Override) - 模拟 Copy / Read / Write



2. 归约式合并 (Reduce) - 模拟 Sum / Max / Min



🧩 算子协议标准图案



算法分析器：内存冲突校验引擎原理

💡 核心洞察

- **全景过程安全防线：** 作为语义校验的严谨补充，专职拦截执行过程中的中间状态破坏与隐藏内存污染隐患。
- **高阶并发裁决机制：** 确立“先判并发偏序，再判地址重叠”的判别准则，精准捕获并发重叠写，彻底消除偶发性 Bug。

🔗 核心算法：三步曲

① 片段化 (Fragmentation)



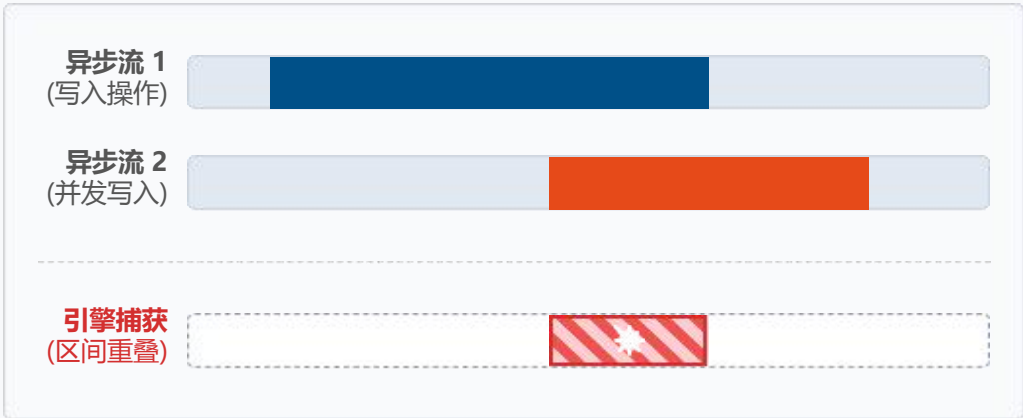
② 并发过滤 (Concurrency Pruning)



工程化原则： 设计优先级为 并发语义正确 > 结果可解释 > 主流规模性能可接受。

⚡ 【步骤 3 放大】并发区间碰撞

仅对保留下来的“并发片段”进行比较。如果两个并发片段访问了重叠的内存区间，且**至少包含一个写操作**，即判定为冲突。



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北极星工具使用介绍

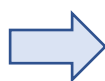
子命令	源文件	职责
log	log_subcmd.cpp	查询/设置日志等级
mock-comm	mock_comm_subcmd.cpp	为单个算子建立通信域
model	model_subcmd.cpp	列出建模文件
plugin	plugin_subcmd.cpp	插件安装/卸载/运行/列表
reset	reset_subcmd.cpp	清空数据库表、重置仿真环境
start	start_subcmd.cpp	启动仿真环境并进入交互 shell
table	table_subcmd.cpp	展示/更新仿真数据库表

plugin	install @checker/@runner	安装checker、runner插件
	uninstall @checker/@runner	卸载checker、runner插件
	run @checker/@runner	执行checker、runner插件

详细安装部署和使用指导见代码仓: https://gitcode.com/cann/hcomm/tree/master/test/hccl_vm

北极星工具实践：常见问题提前拦截

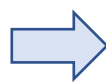
```
work_code > auto_test > HVM_Autotest > result > 20260602100008 > ccu_ms > TC_ccu_ms_128_alltoallv_test_64M_sum_int32 > E runner.log
128690 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:286] [2531771]Current
128691 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_ins_generator_v2.cc:1060] [2531771]CcuInsGeneratorV2::PrepareConstValue Current RepType[36]
128692 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:286] [2531771]Current rep[756] ptr[0x5573263869f0] repType[36]
128693 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_ins_generator_v2.cc:1060] [2531771]CcuInsGeneratorV2::PrepareConstValue Current RepType[36]
128694 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:286] [2531771]Current rep[757] ptr[0x557326386a70] repType[36]
128695 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_ins_generator_v2.cc:1060] [2531771]CcuInsGeneratorV2::PrepareConstValue Current RepType[36]
128696 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:286] [2531771]Current rep[758] ptr[0x557326386af0] repType[36]
128697 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_ins_generator_v2.cc:1060] [2531771]CcuInsGeneratorV2::PrepareConstValue Current RepType[36]
128698 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:286] [2531771]Current rep[759] ptr[0x557326386b70] repType[36]
128699 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_ins_generator_v2.cc:1060] [2531771]CcuInsGeneratorV2::PrepareConstValue Current RepType[36]
128700 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:286] [2531771]Current rep[760] ptr[0x557326386bf0] repType[36]
128701 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_ins_generator_v2.cc:1060] [2531771]CcuInsGeneratorV2::PrepareConstValue Current RepType[36]
128702 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:286] [2531771]Current rep[761] ptr[0x557326386c70] repType[36]
128703 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_ins_generator_v2.cc:1060] [2531771]CcuInsGeneratorV2::PrepareConstValue Current RepType[36]
128704 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:130] [2531771]DumpResReqInfo: dieId[0], msReq[0], blockMsReq[128], ckeReq[32],
128705 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:130] [2531771]DumpResReqInfo: dieId[1], msReq[0], blockMsReq[128], ckeReq[32],
128706 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:164] [2531771]GetResPackTotalResNum:dumpInfos success.
128707 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:340] [2531771]Resource request: dieId[0], ms[0], blockMs[64], cke[1], blockcke[0],
128708 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:130] [2531771]DumpResReqInfo: dieId[0], msReq[0], blockMsReq[64], ckeReq[1], b
128709 [error][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:204] [2531771][CcuKernelMgr][CheckResIfAvailable] dieId[0] not enough, msReq[
128710 [warning][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:295] [2531771][CcuKernelMgr][AllocRes] resource is not enough.
128711 [error][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][ccu_kernel_mgr.cc:109] [2531771][Register]call trace: hcclRet -> 7
128712 [error][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][coll_comm_res_c_adpt.cc:366] [2531771][HcclCcuKernelRegister]call trace: hcclRet -> 7
128713 [error][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][op_common.cc:1521] [2531771][HcclGetCcuKernel]call trace: hcclRet -> 7
128714 [error][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][op_common.cc:1459] [2531771][HcclAllocAlgResourceCcu]call trace: hcclRet -> 7
128715 [warning][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][op_common.cc:1415] [2531771][HcclAllocAlgResourceCcu] resource unavailable, try to fallback.
128716 [error][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][op_common.cc:910] [2531771][HcclGetAlgRes]call trace: hcclRet -> 7
128717 [warning][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][op_common.cc:548] [2531771][HcclGetAlgRes] resource unavailable, try to fallback.
128718 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][hccl_independent_op_ctx.cc:65] [2531771]Entry-HcclEngineCtxCreate:comm[hccl_world_group]
128719 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][engine_ctx.cc:39] [2531771][CreateCommEngineCtx]create context success, tag[CcuAlltoAllVMesh1D_
128720 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][hccl_independent_op_ctx.cc:65] [2531771]HcclEngineCtxCreate success, ctxTag[CcuAlltoAllVMesh1D_
128721 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][op_common.cc:458] [2531771]Start to execute ReSelector.
128722 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][execute_selector.cc:43] [2531771][Algo][Selector] The selector nums of otype[8] is [1].
128723 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][op_common.cc:2208] [2531771]Start CheckHostDPUPOnly
128724 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][rank_graph.cc:143] [2531771][RankGraph][GetLevelNum] validLevelNum[2]
128725
128726 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][hccl_independent_rank_graph.cc:109] [2531771]HcclRankGraphGetLayers success, netLayerNum [2]
128727 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][op_common.cc:2234] [2531771]Skip checking layer[0], topoLevelNums is [2]
128728 [info][PID:2531771][TID:2531771][log_stub.cc][DlogPrintStub] [HCCL_LOG][hccl_independent_rank_graph.cc:109] [2531771]HcclRankGraphGetLayers success, topoLevelNums is [2]
```



模拟运行器报错：
CCU资源不足

北极星工具实践：算法逻辑错误提前识别

```
267, label=original, originalSrcCount=1, originalDstCount=1, mergedSrcCount=1, mergedDstCount=1, dedupMergedSrcCount=1, dedupMergedDstCount=1
[debug][PID:14629][TID:16683][TaskGraphSingleTaskCheckV3.cc][CheckBatchTransSlices] [TaskGraphSingleTaskCheckV3] Batch trans merged slices checked, nodeId=3
302, label=original, originalSrcCount=1, originalDstCount=1, mergedSrcCount=1, mergedDstCount=1, dedupMergedSrcCount=1, dedupMergedDstCount=1
[debug][PID:14629][TID:16683][TaskGraphSingleTaskCheckV3.cc][CheckBatchTransSlices] [TaskGraphSingleTaskCheckV3] Batch trans merged slices checked, nodeId=3
451, label=original, originalSrcCount=1, originalDstCount=1, mergedSrcCount=1, mergedDstCount=1, dedupMergedSrcCount=1, dedupMergedDstCount=1
[debug][PID:14629][TID:16683][TaskGraphSingleTaskCheckV3.cc][CheckBatchTransSlices] [TaskGraphSingleTaskCheckV3] Batch trans merged slices checked, nodeId=3
454, label=original, originalSrcCount=1, originalDstCount=1, mergedSrcCount=1, mergedDstCount=1, dedupMergedSrcCount=1, dedupMergedDstCount=1
[debug][PID:14629][TID:16683][TaskGraphSingleTaskCheckV3.cc][CheckBatchTransSlices] [TaskGraphSingleTaskCheckV3] Batch trans merged slices checked, nodeId=3
492, label=original, originalSrcCount=1, originalDstCount=1, mergedSrcCount=1, mergedDstCount=1, dedupMergedSrcCount=1, dedupMergedDstCount=1
[debug][PID:14629][TID:16683][TaskGraphSingleTaskCheckV3.cc][CheckBatchTransSlices] [TaskGraphSingleTaskCheckV3] Batch trans merged slices checked, nodeId=3
495, label=original, originalSrcCount=1, originalDstCount=1, mergedSrcCount=1, mergedDstCount=1, dedupMergedSrcCount=1, dedupMergedDstCount=1
[debug][PID:14629][TID:16683][TaskGraphSingleTaskCheckV3.cc][CheckBatchTransSlices] [TaskGraphSingleTaskCheckV3] Batch trans merged slices checked, nodeId=3
264, label=original, originalSrcCount=1, originalDstCount=1, mergedSrcCount=1, mergedDstCount=1, dedupMergedSrcCount=1, dedupMergedDstCount=1
[debug][PID:14629][TID:16683][TaskGraphSingleTaskCheckV3.cc][CheckBatchTransSlices] [TaskGraphSingleTaskCheckV3] Batch trans merged slices checked, nodeId=3
102, label=original, originalSrcCount=1, originalDstCount=1, mergedSrcCount=1, mergedDstCount=1, dedupMergedSrcCount=1, dedupMergedDstCount=1
[error][PID:14629][TID:16683][TaskGraphSingleTaskCheckV3.cc][CheckTwoSliceOverlap] [TaskGraphSingleTaskCheckV3] Slice conflict, rank=0, stream=0, node=[Task
TransMem] node=1414, rank=0, stream=0, queue=0, protocol=CCU, src={rank=0, mem=CCL, offset=0x1000, len=0x100}, dst={rank=0, mem=CCL, offset=0x1000, len=0x10
0}, ccuTrace={valid=1, op=ccu, role=local_copy, instrId=155, queueId=0, position=rank=0, stream=0, queue=0}, lhs={rank=0, mem=2, offset=0x1000, len=0x100},
rhs={rank=0, mem=2, offset=0x1000, len=0x100}
[warning][PID:14629][TID:16683][checker.cc][~V3StageTimer] [TaskGraphGeneratorV3][StageTime] CheckGraphV3TaskMem failed, costMs=12, nodeCount=4096, dataTask
NodeCount=338
[error][PID:14629][TID:16683][log_stub.cc][DlogPrintStub] [/home/h60059641_2404/hvm_workspace_0609/CheckerL2/src/plugin/solver/virtual_runtime/checker/src/c
hecker/checker.cc:1490] [16683][GenAndCheckGraphV3]call trace: hccRet -> 4
[warning][PID:14629][TID:16683][main.cc][ProcessOneOpGroup] -----[V3 Checker Finished]-----
[info][PID:14629][TID:16683][main.cc][ProcessOneOpGroup] [ProcessOneOpGroup] new checker finished, ret=4
[info][PID:14629][TID:16683][main.cc][ProcessOneOpGroup] [ProcessOneOpGroup] old checker disabled by manifest setting.
[error][PID:14629][TID:16683][main.cc][RunChecker] [RunChecker] op[0] failed with error code 4
(hvm)$>
```



算法分析器报错：
内存冲突校验不通过

Thank you.

社区愿景：打造开放易用、技术领先的AI算力新生态

社区使命：使能开发者基于CANN社区自主研究创新，构筑根深叶茂、跨产业协同共享共赢的CANN生态

Vision: Building an Open, Easy-to-Use, and Technology-leading AI Computing Ecosystem

Mission: Enable developers to independently research and innovate based on the CANN community and build a win-win CANN ecosystem with deep roots and cross-industry collaboration and sharing.



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